

Name: _____

_____/ 60
Max: 69 pts

1. Which of the following is **NOT** true about **ActionListener**? (2 pts)
 - a. **ActionListener** is an interface in Java
 - b. **ActionListener** must be defined in an anonymous inner class
 - c. **ActionListener** can be implemented in a lambda expression
 - d. **ActionListener** can be used to add functionality to buttons

2. Which layout manager allows the user to add a component to a cardinal direction? (2 pts)
 - a. **FlowLayout**
 - b. **GridLayout**
 - c. **BorderLayout**
 - d. **BoxLayout**

3. Which of the following is **NOT** true about exceptions? (2 pts)
 - a. **Exception** is the base class for all exceptions in Java
 - b. Custom exceptions should inherit from an existing exception class
 - c. Exceptions often include a message string associated with them
 - d. Runtime exceptions must be handled with a try/catch or explicitly thrown

4. Which of the following is used for defining natural object sorting? (2 pts)
 - a. **Comparable**
 - b. **Comparator**
 - c. **ActionListener**
 - d. None of the above

5. Which of the following is not a function of the **super** keyword? (2 pts)
 - a. To call the parent class constructor
 - b. To reference the child of the parent class
 - c. To reference a method of the parent class
 - d. To reference a field of the parent class

Use the code block to answer the following question:

```
1  public static void doThing() {  
2      boolean b = true;  
3      try {  
4          System.out.print("Apple");  
5          if (b) {  
6              throw new Exception();  
7          }  
8          throw new IllegalArgumentException();  
9      } catch (IllegalArgumentException iae) {  
10         System.out.print("Banana");  
11     } catch (Exception e) {  
12         System.out.print("Pear");  
13         return;  
14     } finally {  
15         System.out.print("Orange");  
16     }  
17 }
```

6. Write the output of the `doThing()` method when it is called (1 pts)

7. Match the following to their definitions (7 pts)

_____ implements	a. Denotes that a class is using an interface
_____ JPanel	b. Denotes that a class inherits from another
_____ extends	c. Common Swing Component
_____ hello	d. A class sthat cannot be directly instantiated
_____ Abstract Class	e. world

Use the diagram to answer the following question: (Lines denote inheritance)

8. **Circle all** of the following that are valid statements according to the diagram: (2 pts)

- a. `Animal a = new Animal();`
- b. `Dog d = new Dog();`
- c. `Cat c = new Animal();`
- d. `Cat c = new Dog();`
- e. `Animal a = new Cat();`

9. Abstract classes, interfaces, and multi-layered applications are all examples of (2 pts)

- a. Separation of Concern
- b. Dynamic Dispatch
- c. Inheritance
- d. **ActionListener** Interfaces

10. Which of the following will **always** cause a `ClassCastException` to occur? (2 pts)

- a. Casting a child class to a parent class
- b. Casting a parent class to a child class
- c. Casting between unrelated classes
- d. Casting between primitive types

11. Select all that are true about using `Runnable` interface: (4 pts)

- a. It can be implemented via a lambda expression
- b. It allows for a class to extend another class and be multi-threaded
- c. It requires the `start()` method to be overridden
- d. It must be used with a class that extends `Thread`

CS-1181 Quiz #1

Use the `Student` class to answer the following questions:

```
1  class Student {
2      private String name;
3
4      public Student(String name){
5          this.name = name;
6      }
7      public void setName(String newName){
8          this.name = newName;
9      }
10
11     public String toString(){
12         return this.name;
13     }
14 }
```

```
1  public static void main(String[] args) {
2      Student s1 = new Student("Alice");
3      Student s2 = s1;
4      s2.setName("Bob");
5      System.out.println(s1);
6      System.out.println(s2);
7  }
```

12. What is the output of the code? (2 pts)

- a. Alice
Bob
- b. Alice
Alice
- c. Bob
Alice
- d. Bob
Bob

13. Why does this issue occur? (2 pts)

- a. Because a deep copy is performed
- b. Because inheritance is used
- c. Because a shallow copy is performed
- d. None of the above

14. Write one line of code to create a deep copy of `s1`: (1 pts)

CS-1181 Quiz #1

15. **BoxLayout** is the default component layout manager (2 pts)

- a. True
- b. False

16. Briefly describe the purpose of the **JPanel** class in Swing? (1 pts)

Use the code block to answer the following question:

```
public class CustomFrame extends JFrame {  
    public CustomFrame() {  
        this.setSize(500, 500);  
        JButton btn1 = new JButton("Submit");  
        this.add(btn1);  
        this.setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);  
        JLabel label = new JLabel("This is some text");  
        JButton btn2 = new JButton("Next");  
        this.add(btn2);  
        this.setVisible(true);  
    }  
}
```

17. What will render when an instance of `CustomFrame` is created? (2 pts)
- a. A frame containing only the "Submit" button
 - b. A frame containing the "Submit" button, "Next" button, and a JLabel
 - c. A frame containing only the "Next" button
 - d. The frame will not render
18. Briefly explain why you chose that answer: (2 pts)

19. Which of the following is **NOT** true about interfaces? (2 pts)
- a. A class in Java can only inherit a single interface
 - b. An interface ensures a class must implement select methods
 - c. You can pass interfaces into methods as data types
 - d. You can declare the left side of variables as an interface

Use the code block to answer the following questions:

```
1  public abstract class Worker {
2      public abstract void goToWork();
3
4      public void performTask(){
5          System.out.println("Performs task...");
6      }
7  }
8
9  public interface TimeTrackable {
10     public void clockIn(String time);
11
12     public void clockOut(String time);
13 }
14
15 public class Intern extends Worker implements TimeTrackable {
16     //...
17 }
18
19 public class Driver {
20
21     public static void main(String[] args) {
22         Intern i = new Intern();
23         performHourlyLabor(i);
24     }
25
26     public static void performHourlyLabor(TimeTrackable t){
27         t.clockIn("9am");
28         t.clockOut("5pm");
29     }
30
31 }
```

20. Circle **all** of the following that are true about the class Intern? (2 pts)

- a. Intern **must** write the body for the clockOut method
- b. Intern **must** write the body for the performTask method
- c. Intern **must** override the body for the goToWork method
- d. Intern **must** override the toString method

21. The call to the method `performHourlyLabor` is an example of: (2 pts)

- a. Dynamic Dispatch
- b. Inheritance
- c. Multi-Layered Application
- d. Type Casting

22. Which method is used for defining custom graphics in Swing? (2 pts)

- a. `setGraphics(Graphics g)`
- b. `draw()`
- c. `addActionListener()`
- d. `paintComponent(Graphics g)`

23. What does the **throws** keyword do? (2 pts)

- a. Indicates that a method could cause an error
- b. Creates and instantiates an exception
- c. Creates a custom exception type
- d. Catches and handles an exception

24. Which of the following is **NOT** true about abstract classes? (2 pts)

- a. If a class has one abstract method, the class must be marked as abstract
- b. Abstract classes can be directly instantiated
- c. Abstract methods must be implemented in a child class
- d. Abstract classes can contain methods with no bodies

25. Write the header for a class that inherits from **Number** and can be sorted in a list: (1 pts)

26. All Java Swing events are handled on the Main Thread (1 pts)

- a. True
- b. False

27. **Circle all** of the following that are concerns when programming GUIs in Swing? (2 pts)

- a. Doing lots of work on one component may block the EDT
- b. Avoiding **OutOfFrameException** when animating components
- c. Adding more than ten components to a JPanel will cause a crash
- d. Adding multiple components **directly** to a JFrame

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- 28.** In the space below, write a generic bubble sort method. At a minimum, the method should:
- Take in an `ArrayList` (or similar) of generic elements that are comparable to themselves
 - Sort the elements using the bubble sort algorithm
 - You are welcome to either modify the list passed in or return a sorted version of the list
 - Please ensure that the method can sort an `ArrayList` of any length (8 pts)

- 29.** Mark the following statements as either True (T) or False (F): (5 pts)

- I. `LinkedList` utilizes nodes and next pointers to store information _____
- II. `ArrayList` is always faster than `LinkedList` _____
- III. `ArrayList` has more flexible memory usage than `LinkedList` _____
- IV. `ArrayList` and `LinkedList` implement the same ADT _____
- V. `ArrayLists` are faster than `LinkedLists` at retrieval operations _____